DEEP LEARNING LABORATORY

EXPERIMENT 9

MULTILAYER PERCEPTRON IMPLEMENTATION

Aim

To implement a Multilayer Perceptron (MLP) using TensorFlow/Keras, train it on the XOR dataset, and evaluate its predictions and training loss.

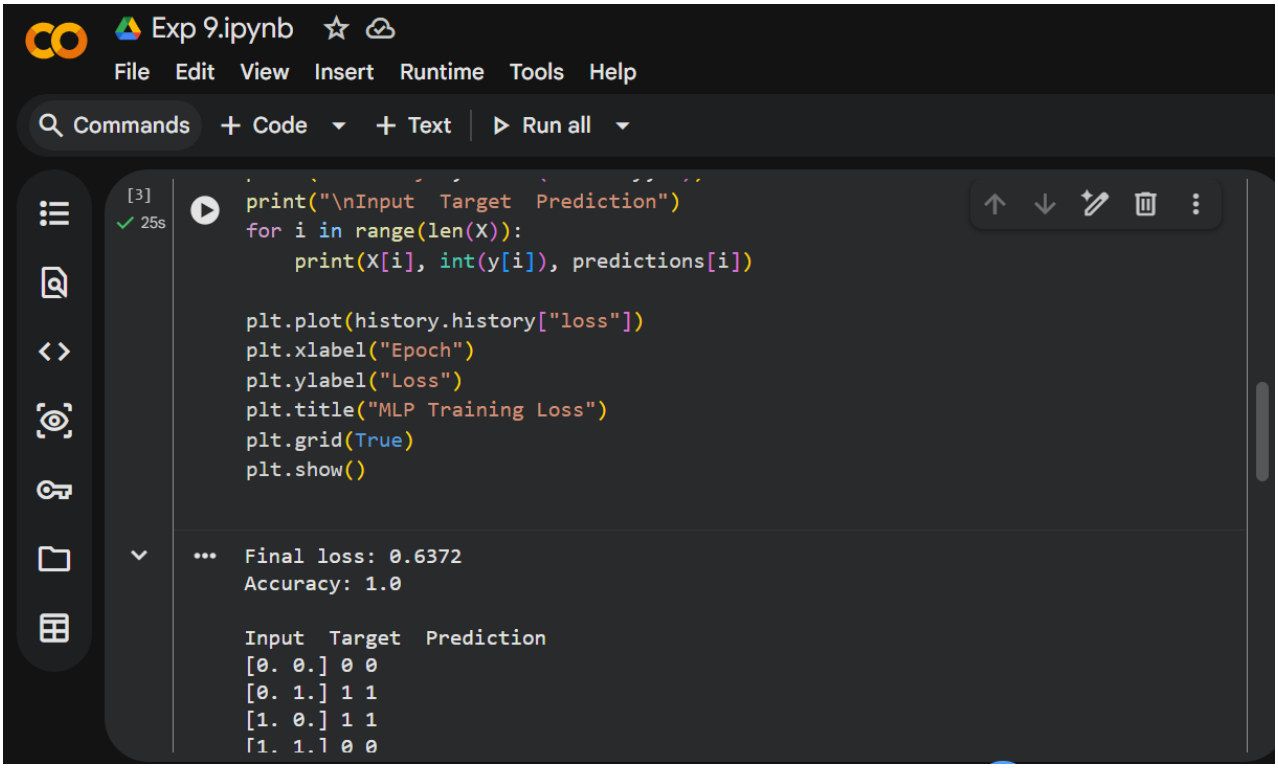
Output

The model reports a final loss of 0.6372 and an accuracy of 1.0. The printed predictions match all four XOR target labels: [0,0] → 0, [0,1] → 1, [1,0] → 1, and [1,1] → 0.

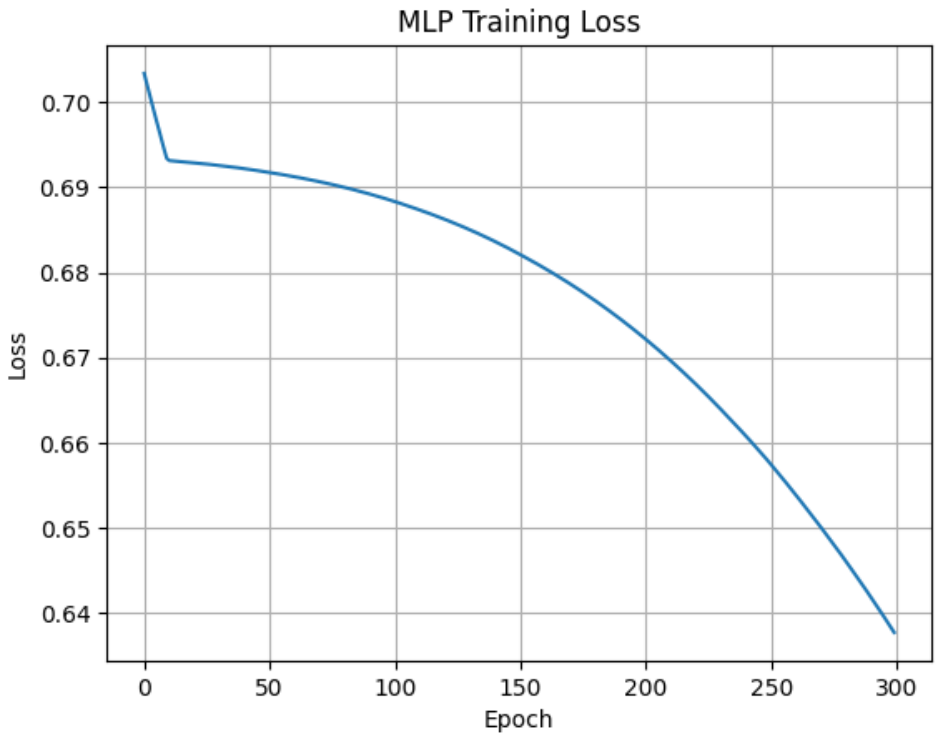
Observed output table

Input	Target	Prediction
[0, 0]	0	0
[0, 1]	1	1
[1, 0]	1	1
[1, 1]	0	0

Colab output screenshot



Training-loss graph


MLP Training Loss
Loss
0.70
0.69
0.68
0.67
0.66
0.65
0.64
0
50
100
150
200
250
300
Epoch